Ibna Kowsar

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Research Interests: Mechanistic interpretability (sparse autoencoders, attribution), parameter-efficient fine-tuning, continual learning, and transfer learning for large language / vision-language models: why models fail during adaptation, and how to repair.

Education

University of California, Davis — California, USA
Ph.D. in Computer Science — *Sep 2025 – Jun 2030 (expected)*

Tennessee State University — Tennessee, USA
M.Sc. in Computer Science, Data Science (GPA: 4.0/4.0); Thesis: attention-based imputation — *Aug 2023 – May 2025*

Experience

Graduate Researcher — Sep 2025 – Present
University of California, Davis; advised by Prof. Xin Liu

- Designed **Concept-Guided Retention (CGR)**, a plug-and-play regularizer that stops fine-tuning from erasing what a model already knows. A **frozen sparse autoencoder** names the concepts inside the model; CGR sorts them into *task* and *keep* sets, then holds the adapter's update inside the task concepts while pinning the kept concepts to their original activations.
- Showed that limiting *where* the weight update lands is not enough on its own, and derived bounds on how far a kept concept can still drift. Earlier PEFT methods constrain the weights but leave activations free.
- Validated across **both vision and language**: **10 vision tasks** (CLIP ViT-B/32) and instruction tuning of **Llama-3.1-8B**, each against **six adapters**. Retention improved for every adapter at comparable accuracy (vision **+2–7%**, LoRA 74.1→80.2; world knowledge **+2–8%**, LoRA TriviaQA 27.6→43.5), and the protected directions stay **human-readable, named concepts**.

Graduate Research Assistant / Research Associate — Aug 2023 – Aug 2025
CIDA Lab, Tennessee State University

- Proposed **DeepIFSAC**, a between-feature/between-sample **attention** framework with contrastive learning; **outperformed 9 SOTA imputers** across MCAR/MAR/MNAR regimes and 10–50% missing rates *(Knowledge-Based Systems, 2025)*.
- Proposed **LATTLE**: transplanted frozen **key/value projections** from a source-finetuned **DistilGPT2** into a gated FT-Transformer, transferring across tabular domains with **no shared features**; **best of 13 methods**, over baselines pretrained on **8B rows** *(arXiv:2511.06161)*.
- Aligned **cluster distributions** jointly with source-target adaptation: accuracy up on **9 of 10 benchmarks**, to **+5.8%** over class-agnostic alignment *(IEEE ICMI; ICISPC)*.

Machine Learning Engineer (OCR Detection & Segmentation) — Jul 2021 – Jan 2023
Apurba Technologies Inc. — *Sunnyvale, CA (remote)*

- Built deep-learning models on the **OCR detection & segmentation** team, advancing Bengali character detection and document layout segmentation with a multi-headed CNN *(IEEE Big Data 2021; ICPR 2022)*.
- Owned the **model deployment pipeline**: containerized models with Docker, served them via scalable Flask/FastAPI inference APIs, cutting inference time by **30%**.

Selected Publications

- **Kowsar, I.**, Medri, I., Meng, F., & Liu, X. (2026). *Confine to the Task, Anchor the Rest: Concept-Guided Retention Regularizer for Parameter-Efficient Fine-Tuning*. **arXiv:2610.17713**.
- **Kowsar, I.**, Rezaei, S., Meng, F., & Liu, X. (2026). *A Benchmark Study on the Reliability of Explainability Methods*. **CVPR 2026 Workshops**, Explainable AI for Computer Vision (**XAI4CV**), 4037–4046.
- Akhter, K. F. B., **Kowsar, I.**, & Samad, M. D. (2026). *Generalized Context in Cross Attention for Transfer Learning of Disjoint Tabular Data*. **Neural Networks**, 109568.
- **Kowsar, I.**, Rabbani, S. B., Hou, Y., & Samad, M. D. (2025). *DeepIFSAC: Deep Imputation Using Feature and Sample Attention within a Contrastive Framework*. **Knowledge-Based Systems**, 318, 113506.
- **Kowsar, I.**, Rabbani, S. B., & Samad, M. D. (2024). *Attention-based Imputation of Missing Values in EHR Tabular Data*. IEEE ICHI. *(24 citations, most-cited first-author work)*

Technical Skills

Interpretability & LLMs: Sparse autoencoders, TransformerLens, SAELens, Captum; activation patching, attribution, probing, model steering; LoRA/QLoRA, vLLM

Deep Learning & Training: Python, Bash, SQL; PyTorch, Hugging Face (Transformers, PEFT, Accelerate), distributed training (DeepSpeed, FSDP), multi-GPU, FlashAttention

ML, Vision & Infra: Scikit-learn, OpenCV, OpenCLIP, einops, contrastive & representation learning, domain adaptation, knowledge distillation; SLURM, Ray, Docker, Weights & Biases, Hydra, AWS (EC2/S3), GCP

Fellowships, Teaching & Service

GGCS Ignite Fellow, UC Davis (2025–2027). Two years of full-time research funding from the CS Graduate Group, with no teaching or service obligation.
Teaching: Graduate TA, UC Davis (2026), ECS 193A/B Senior Capstone; Lecturer, BRAC University (2021–2023).
Service: Reviewer, AMIA Annual Symposium Workshop and IEEE ICHI (2025).